Bluebell and X-CSL lights updater for XP12

Inhaltsverzeichnis

What is it?....2
Quickstart tl;dr....3
Prerequisites:....3
Lets start!....3
Important notes!....3
Usage on the command line....4
Usage of lights_updater.py with parameters explained....4
-h or -- help: show help page....4
-v or -- version: show program version....4
-p or -- path: set path with argument....4
-f or --flashing-beacons: Creates flashing beacons....4
-u or -- undo: undo previous conversion....4
-r or -- remove-backups: remove backup files....4
Redo (with new or changed light params)?....4
Typical command line examples:....5
The GUI usage....6
Menus of lights_updater_gui.py explained....6
File → Exit:....6
Converter → Run standard conversion....6
Converter → Run conversion with flashing beacons....6
Converter → Run undo....6
Converted → Remove backup files! Be careful!....6
CSL → Select CSL directory....6
Help....6
The GUI binaries....7
Linux:....7
Mac OS X:....7
Windows:....7
How long does it take to convert?....8
Specials....8
aircrafts.json....8
light_params.json....8
Custom CSL aircraft:....8

© 2025 Richard Muller

What is it?

When moving on to XP 12 I realized that the somewhat older CSL models from Bluebell and X-CSL lack lights during daylight. Now, with some updates of XP 12, it is partially better, but still not as good as the XP 12 native lighting can be.

So I wrote this program to convert those CSL models to the new XP 12 lighting.

During the development of this program I added some extra tuning (flashing beacons, varying flash sequences for beacons and strobes).

© 2025 Richard Muller

Quickstart tl;dr

- **Make a backup of your CSL directories prior to changing your files!**
- **Some custom CSL objects will end up with no lights at all or only some lights are available and others don’t. Caused by incomplete or inappropriate light params which get removed in the conversion process.**

Prerequisites:

Python must be installed. Following is tested:

Python 3.10 and 3.12 on Linux

Python 3.12 on Windows 10

Python 3.13 on Windows 11

Python 3.13 on MacOS Big Sur (couldn’t test on newer ones)

For Bluebell CSL packages make sure you have the version from x-plane.org.

Lets start!

1. Unzip the file to a place of choice
2. Edit the configs/config.ini and set the csl_path value. See notices!
3. Start the tool with ./lights_updater.py or python ./lights_updater.py
4. Start the tool with ./lights_updater.py -h for help on the command line parameters and / or in case you don’t use the config file.

Important notes!

- **The path must contain at least one xsb_aircraft.txt.**
- **Path in config.json and on the command line must be in quotation marks!**
- **Path in config.json must be with forward slashes (i.e. C:/MyPath/CSLs/), even on Windows! Backslashes cause troubles!**
- **Python prior to version 3.10 will throw an error.**

© 2025 Richard Muller

Usage on the command line

Usage of lights_updater.py with parameters explained

-h or -- help: show help page.

-v or -- version: show program version

-p or -- path: set path with argument

You can specify a path besides the setting in the config.ini with **-p** or **--path** and the path to your csl file in quotes.

For example: ./lights-updater.py -p “D:/X-Plane 12/Resources/etc/etc”

- **Path on the command line MUST have quotes if there are (white-)spaces in the path!!**
- **Path on Windows should also be with forward slashes.**

-f or --flashing-beacons: Creates flashing beacons

Nowadays, the bigger airplanes have flashing beacons. This will simulate that.

-u or -- undo: undo previous conversion

This tool makes a backup of each object file that is in the xsb_aircrafts.txt, so there is an option to undo the changes with the **-u,** or **--undo** option.

-r or -- remove-backups: remove backup files

If all is good, there is a option to delete the backups after all is fine. This can be done by starting the tool with the **-r** or --**remove-backups** option.

- **But be careful. Gone is gone. At least on Linux.**

Redo (with new or changed light params)?

You can restart the tool if you change values in the light_params.json.

- **But! Switching between normal beacons and flashing will not work.**
- **Recommended workflow is always to first run undo and run the conversion.**

© 2025 Richard Muller

Typical command line examples:

A typical command line to process CSL files will look like this:

python ./lights_updater.py -p “C:/X-Plane 12/Resources/plugins/LiveTraffic/Resources/CSL”

With the undo function:

python ./lights_updater.py -p “C:/X-Plane 12/Resources/plugins/LiveTraffic/Resources/CSL” -u

With flashing beacons:

Python ./lights_updater.py -p “C:/X-Plane 12/Resources/plugins/LiveTraffic/Resources/CSL” -f

© 2025 Richard Muller

The GUI usage

Just run python lights_updater_gui.py, double click it or on Mac OS X use start with Python Launcher.

Menus of lights_updater_gui.py explained

File → Exit:

Closes the gui

Converter → Run standard conversion

Converts the lights without changing their behavior.

Converter → Run conversion with flashing beacons

Converts the lights and where applicable converts beacons to flashing beacons.

There is a difference in frequency between Airbus and other aircraft.

Converter → Run undo

Reverts all aircraft objects back to the previously made backups. Removing the backups.

Converted → Remove backup files! Be careful!

Removes the backup files! If you need more disk space or you don’t want to keep them.

Be careful with this option. Gone is gone!

CSL → Select CSL directory

Select the directory where the CSL models are stored. Uses the OS specific file picker app.

Help

Some information.

© 2025 Richard Muller

The GUI binaries

Should all be starting with doubleclick.

Linux:

LIN_lights_updater_gui

Mac OS X:

MAC_lights_updater_gui

Only tested on BigSur. I don’t have access to another version.

Windows:

WIN_lights_updater_gui.exe

Tested on Win 10 and Win 11.

© 2025 Richard Muller

How long does it take to convert?

Times measure on my rig with SSD with X-Plane 12:

INFO : Processing done, 2198 files have been processed!
INFO : It took 16.81 seconds to complete lights_updater

On an older HDD:

INFO : Processing done, 2198 files have been processed!
INFO : It took 53.83 seconds to complete lights_updater

Your millage will vary. Hardware, OS and a lot more possibilities :-)

Specials

aircrafts.json

This file contains the aircraft's by type as far as I could determine the correct type.
Data is bases on Doc8643 from ICAO.
Also which aircraft have flashing beacons is specified here.

If you detect an aircraft not being converted correctly or is missing completely, you can add it that file.

light_params.json

This file contains the lightparams per aircraft category. Those are partially copied from airplanes that are XP12 compatible, partially an estimation by me.
Here you can change the params for each light.

Custom CSL aircraft:

Supports now many Custom CSL aircraft. Some still are bad guys who misbehave :-)

I did some tests. Not all were successful.
Sometimes the xsb_aircraft.txt is not correct filled with the aircraft data (case-sensitive filenames differ from entry in xsb_aircraft.txt) or aircraft are completely missing. So some will work, some don’t. Try if you dare :-) The rate of success could be higher on Windows. As long as the xsb_aircraft.txt is correct.

© 2025 Richard Muller